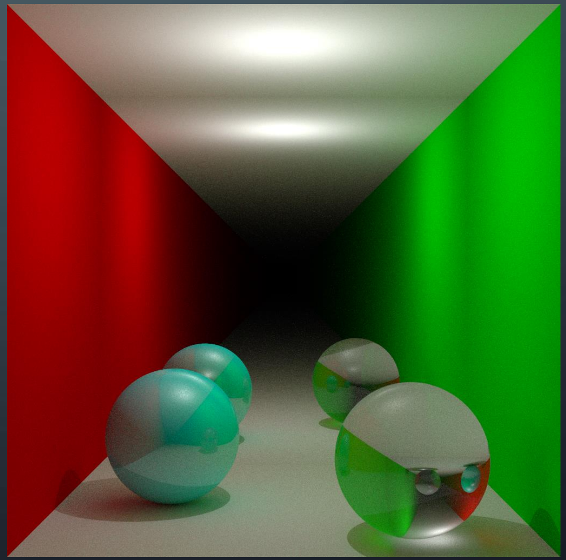
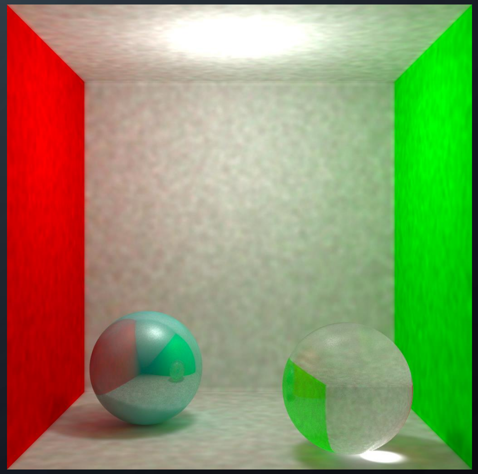
INFORMÁTICA GRÁFICA (2024-2025)

ALEX PETROV (NIP: 81209)





ÍNDICE

- Desarrollo del trabajo
- Path-Tracer
  - Desarrollo
  - Geometrías
  - Paralelizacion
- Photon-Mapper
  - Desarrollo
  - Kernels adicionales
- Comparaciones

DESARROLLO DEL TRABAJO

- Lenguaje: C++
  - Rapidez
  - Diseño orientado a objetos
  - Paralelización

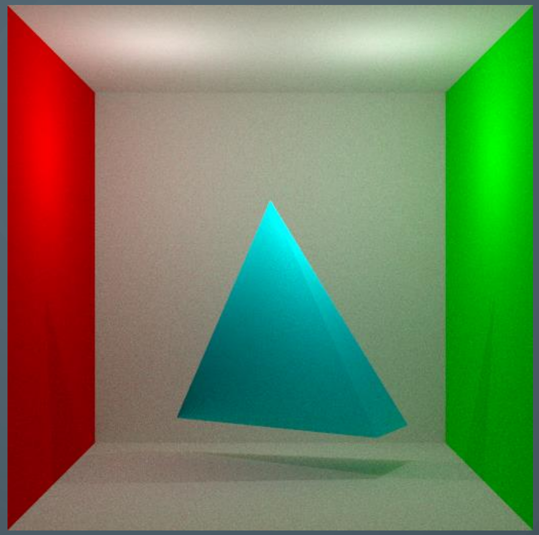
PATH-TRACER: DESARROLLO

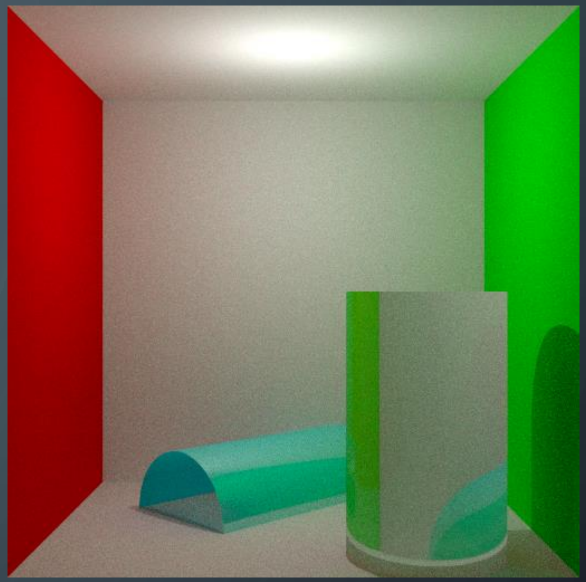
- Ajustes
  - Numero de rebotes de cada path
  - Numero de paths por intersección
    (por defecto a 1, ya que es muy costoso)
  - Numero de rayos por pixel
  - Dimensión de la imagen

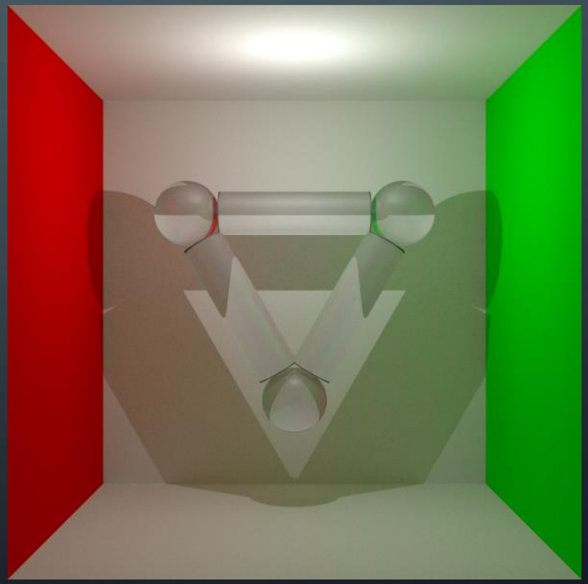
```
/* SETTINGS */

const size_t MAX_BOUNCES = 6;
const size_t MAX_PATHS = 1;

const size_t MAX_RAYS_PER_PIXEL = 64;
const size_t IMAGE_WIDTH = 512;
const size_t IMAGE_HEIGHT = 512;
```

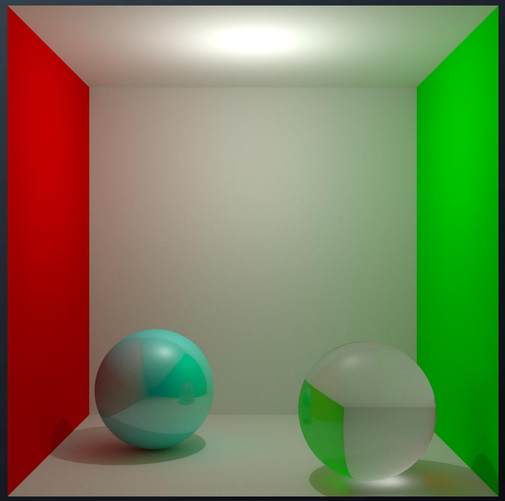
PATH-TRACER: GEOMETRÍAS

- Distintas figuras:
  - Planos
  - Esferas
  - Cilindros
  - Triángulos
  - Mallas de triángulos



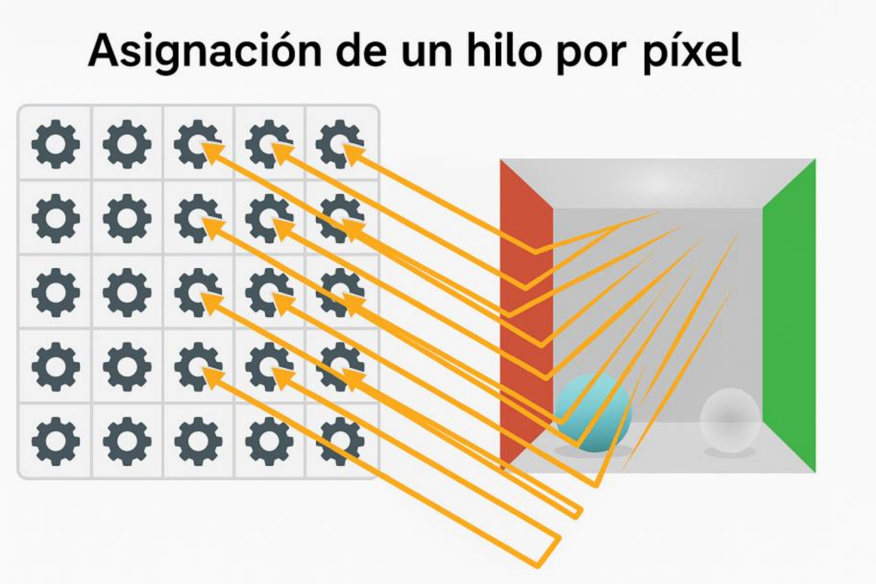






PATH-TRACER: PARALELIZACION

- Pool de threads para ahorrar tiempo al crear threads
- Thread separado para una barra de progresión
- Temporizador automático




```
[PhotonMap Generation Timer] Time: 3.39442s
[=============================================] 100%
[Render Timer] Time: 20m 9.6634s
```

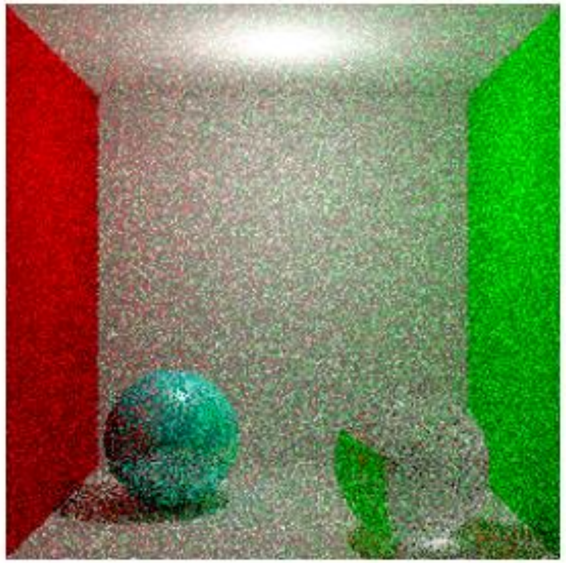


Figura 3.2 - 2 muestras/pixel

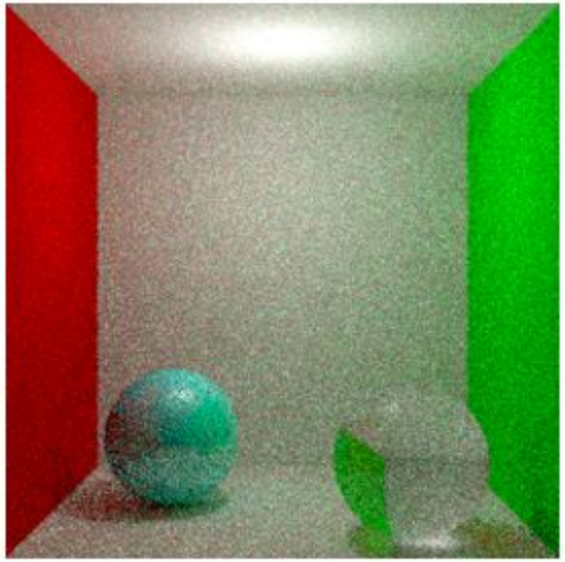


Figura 3.2 - 8 muestras/pixel

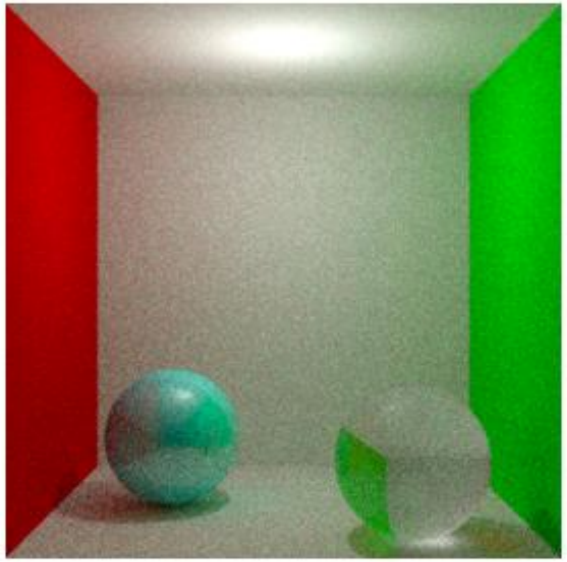


Figura 3.3 - 32 muestras/pixel

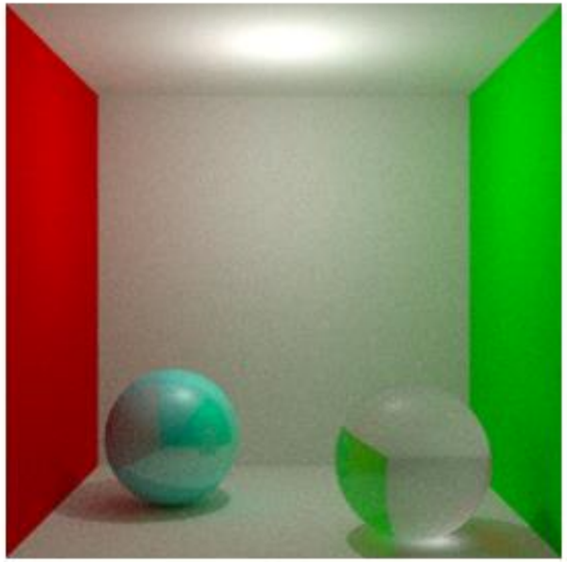


Figura 3.4 - 128 muestras/pixel

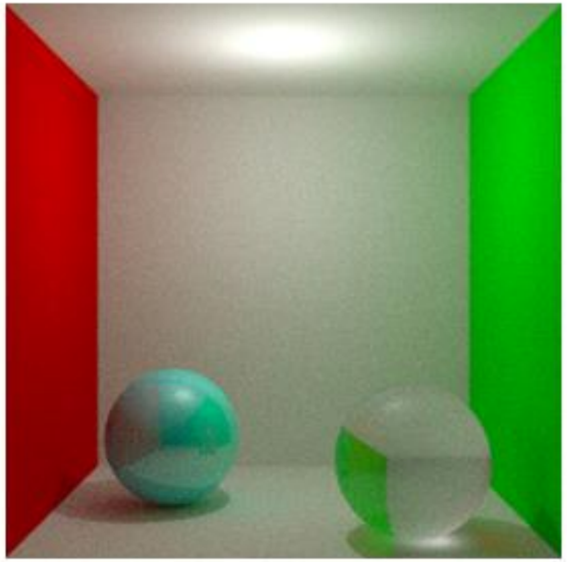


Figura 3.5 - 512 muestras/pixel

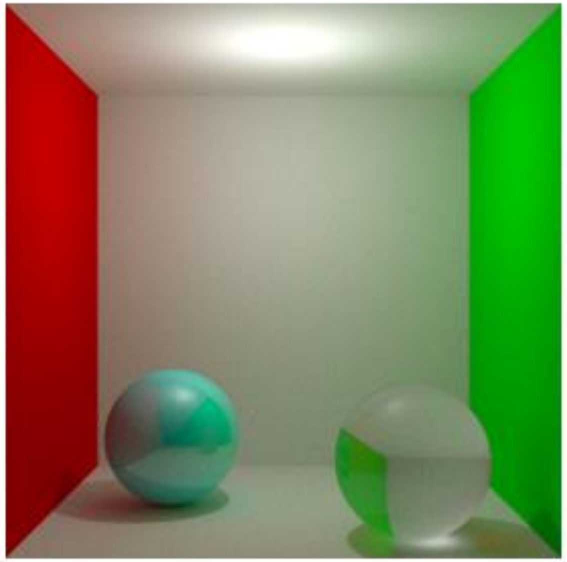


Figura 3.6 - 2048 muestras/pixel

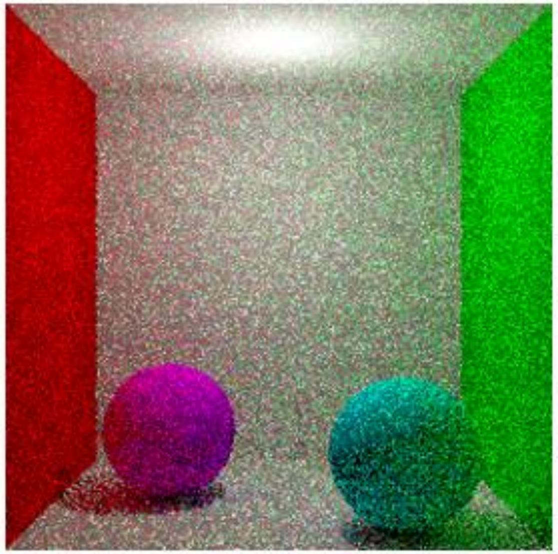


Figura 3.3 - 2 muestras/pixel

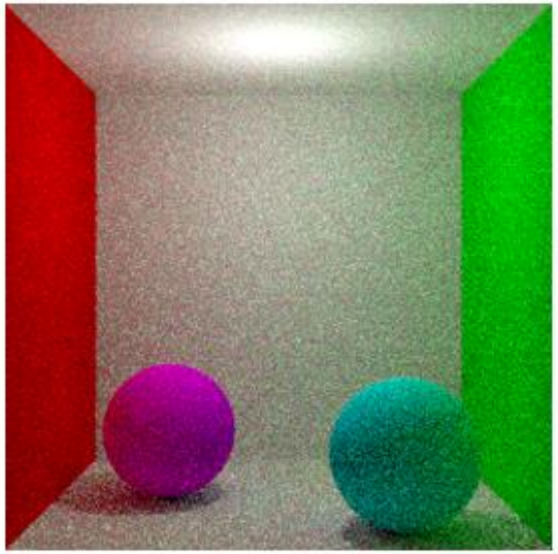


Figura 3.8 - 8 muestras/pixel

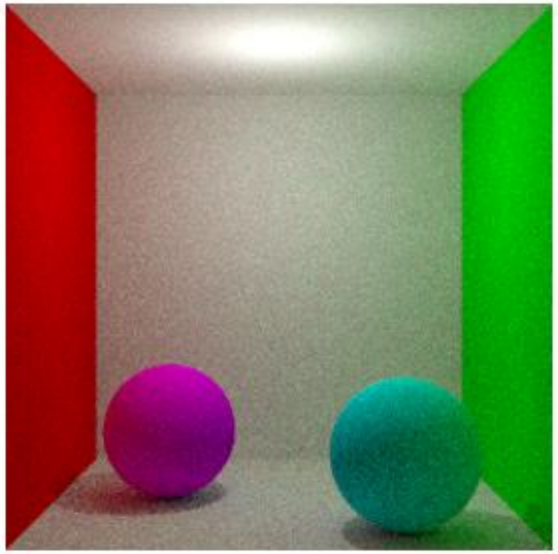


Figura 3.9 - 32 muestras/pixel

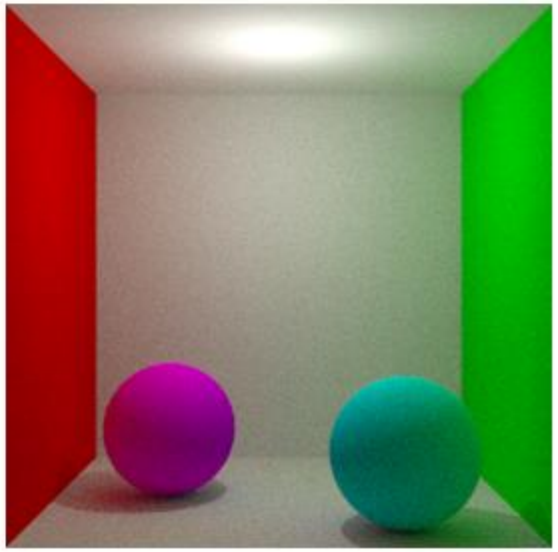


Figura 3.10 - 128 muestras/pixel

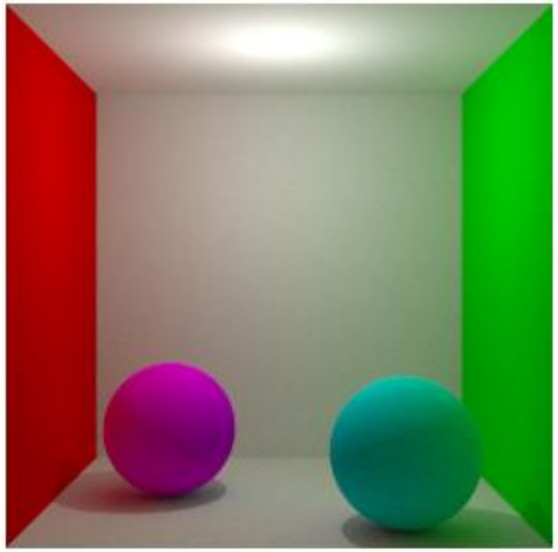
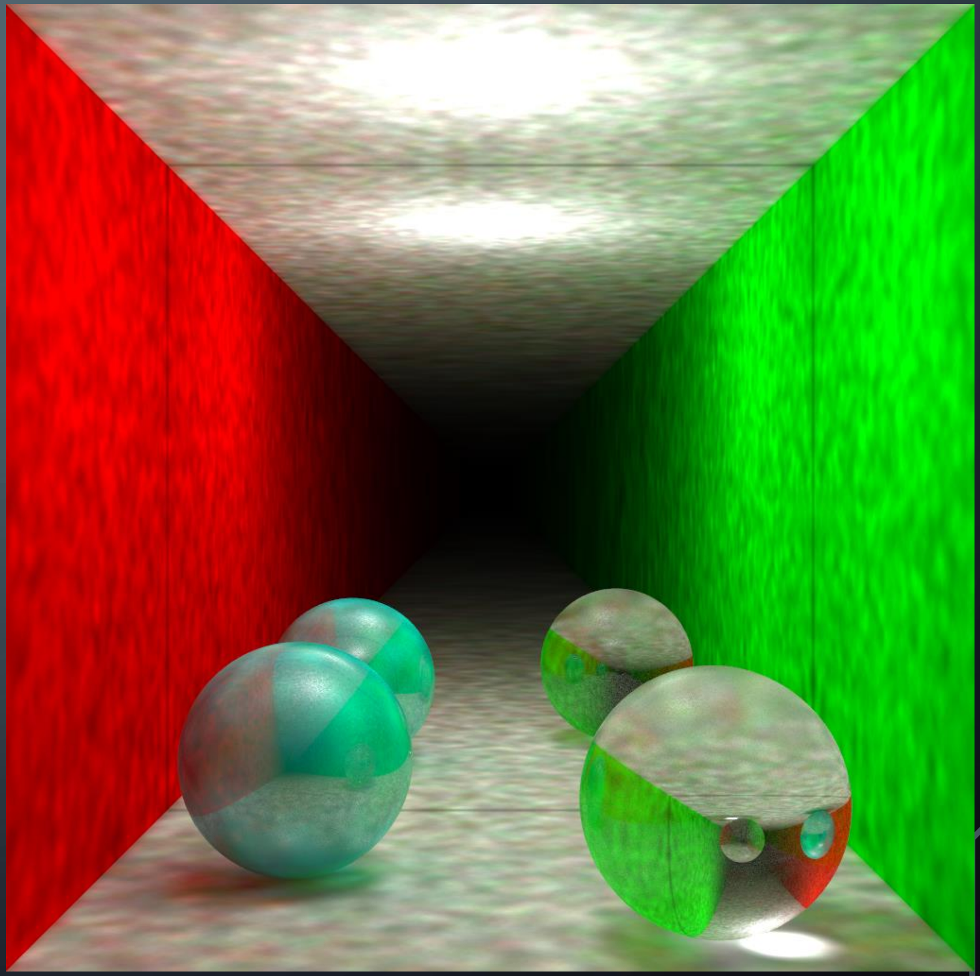


Figura 3.11 - 512 muestras/pixel

Figura 3.12 - 2024 muestras/pixel

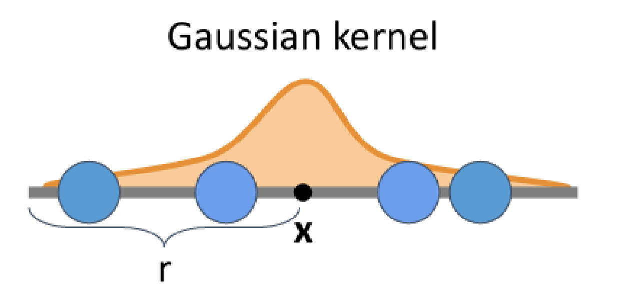
PHOTON-MAPPER: DESARROLLO

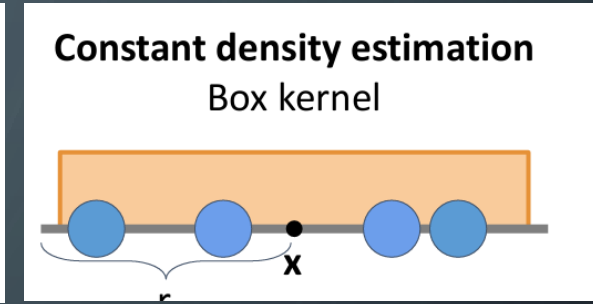
- Emisión y almacenamiento: Se emiten fotones desde las fuentes de luz y se almacenan sus colisiones en una estructura KD-Tree.
- Estimación de radiantica: En fase de rende rizado, se busca la contribución de los fotones cercanos a un punto visible para estimar la luz indirecta.



PHOTON-MAPPER: KERNELS ADICIONALES

- Kernel gaussiano:
  - Permite dar un peso mayor a los fotones mas cercanos





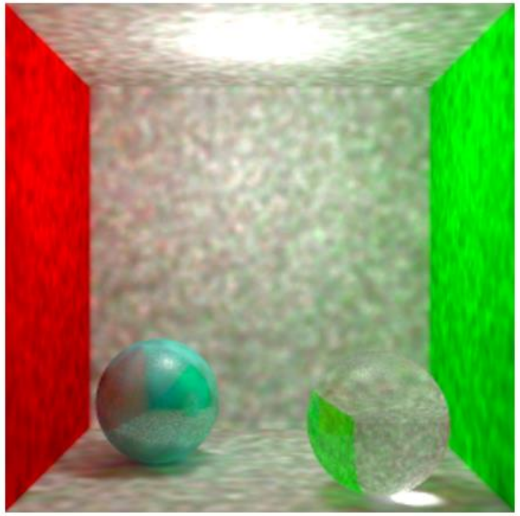


Figura 4.2 - 100K fotones. 100 vecinos. kernel gaussiano

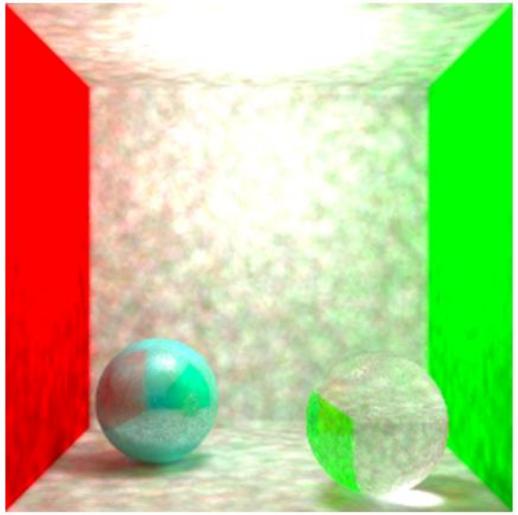
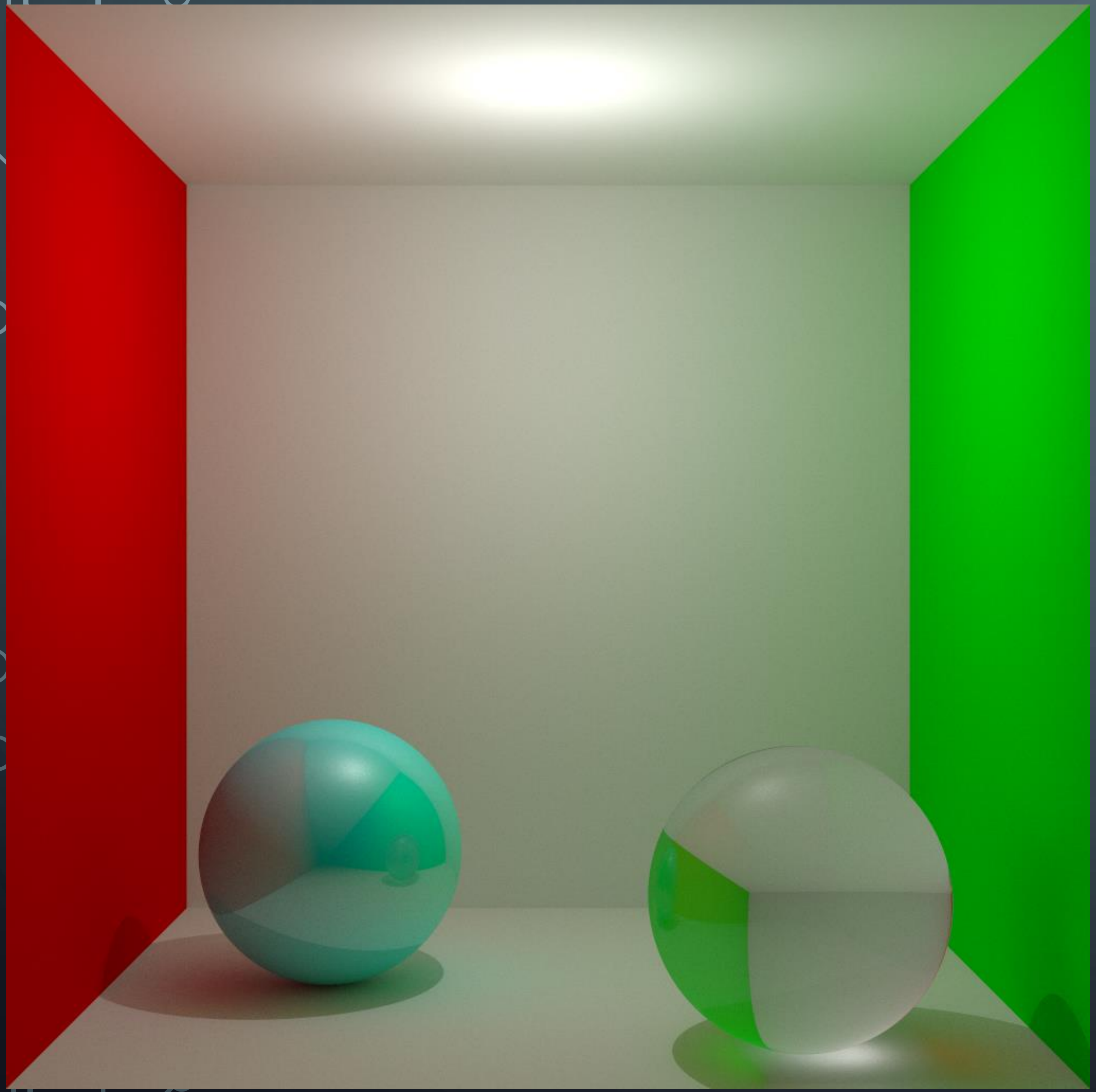
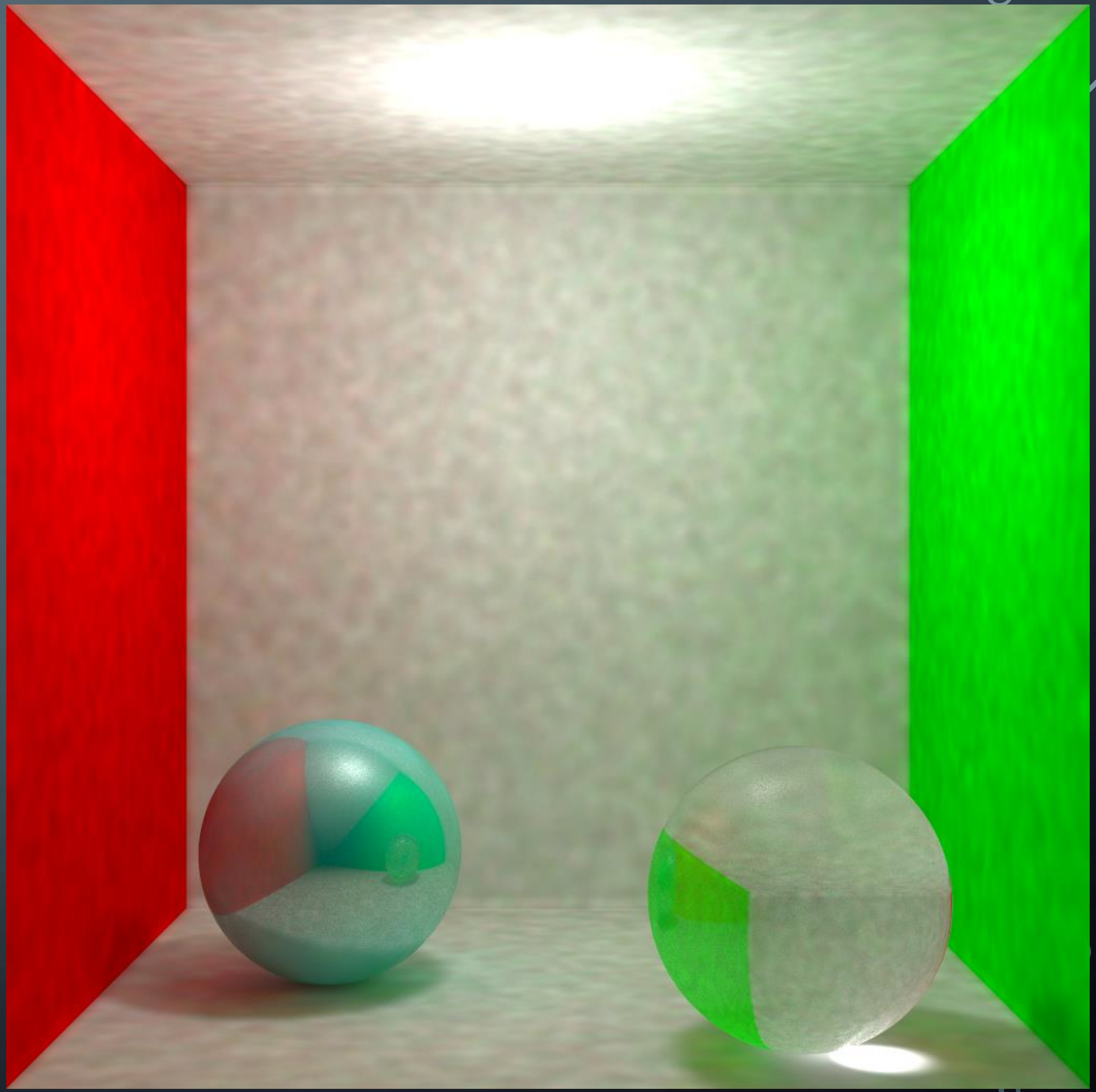
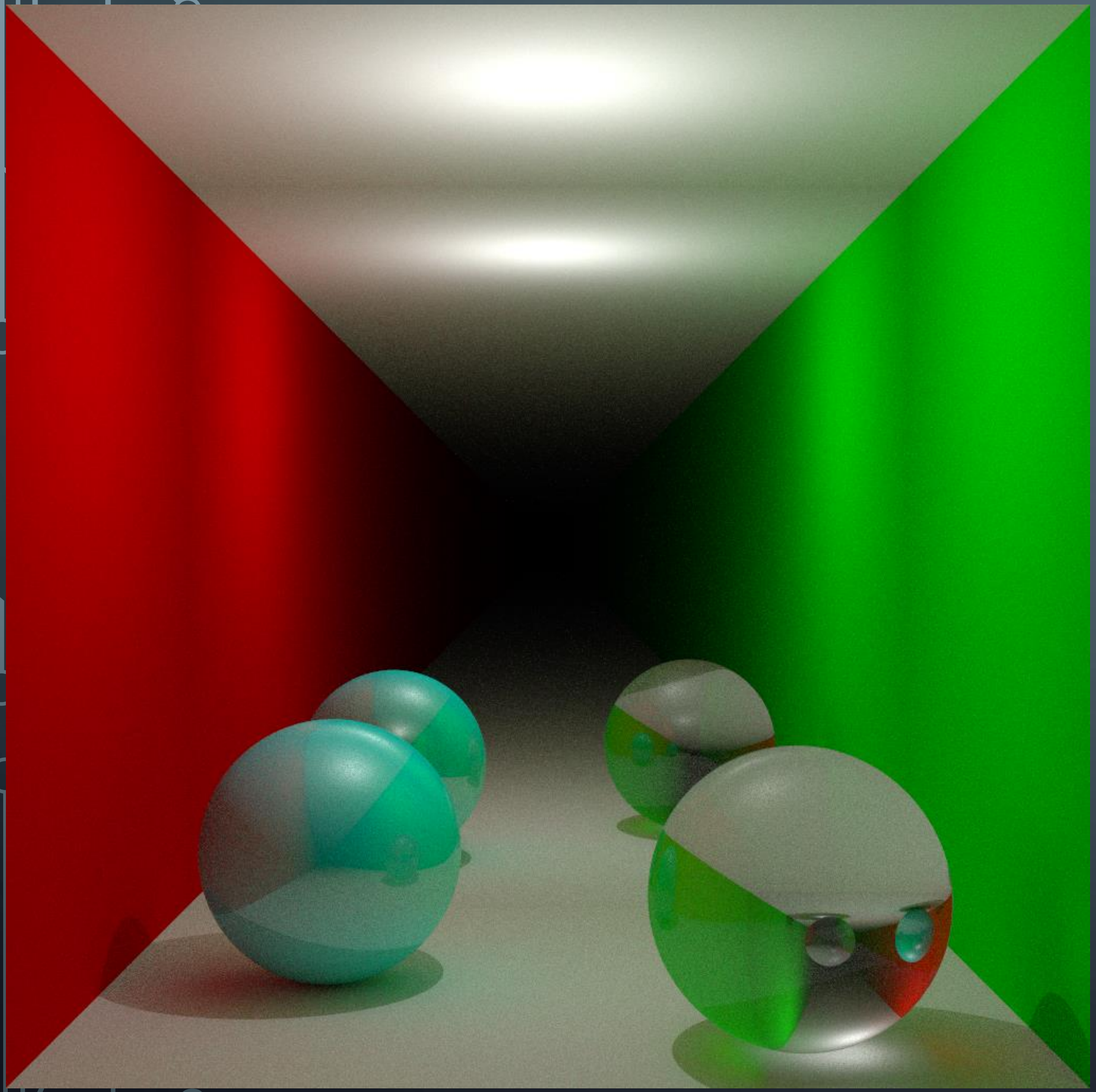
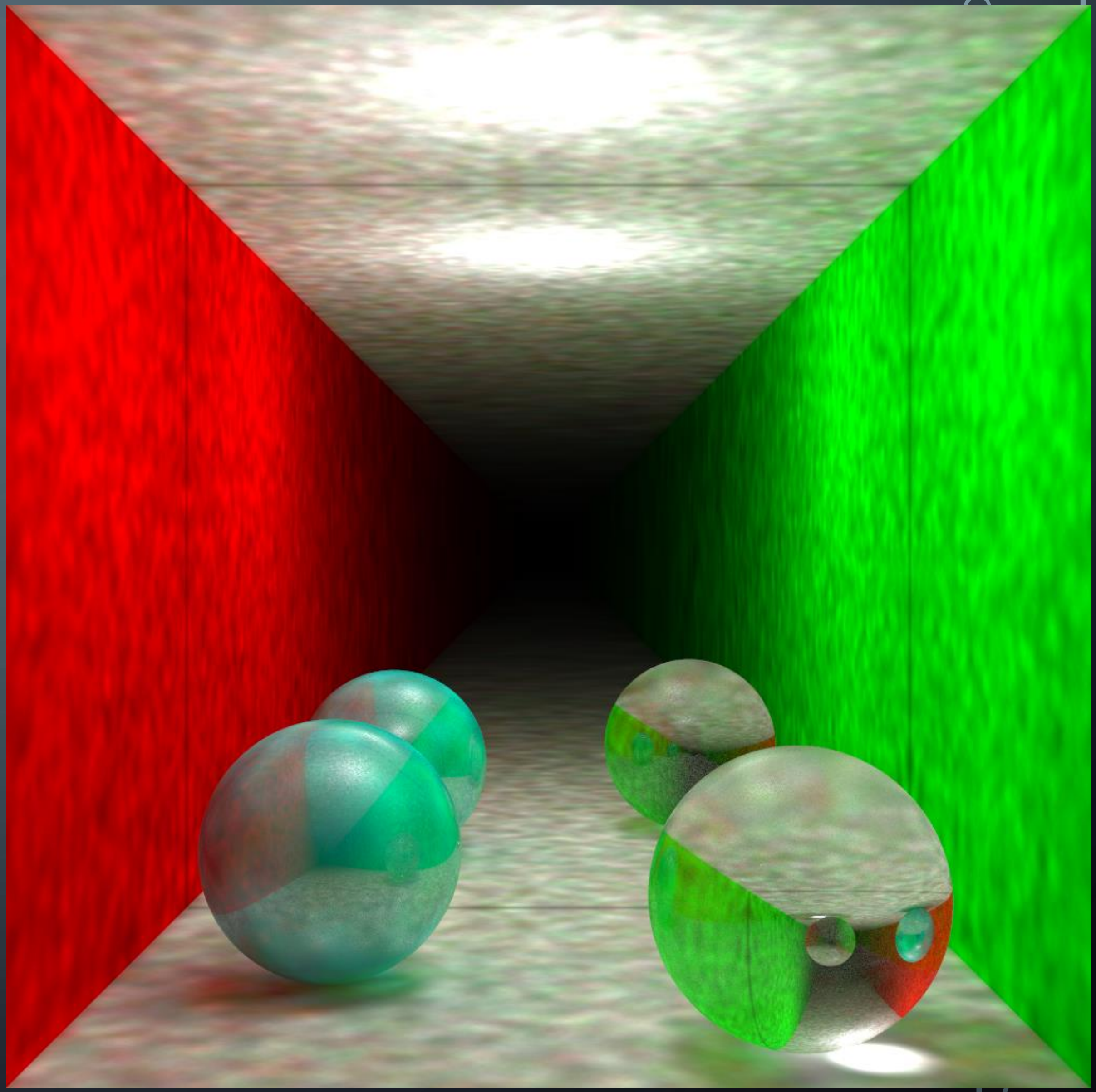


Figura 4.2 - 100K fotones. 100 vecinos. Kernel constante





FIN